Chapter 12.

12.2. (a) $\det\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} = 2\cdot 4 - (-1)\cdot(-3) = \boxed{5}$

(b) $\det\begin{bmatrix} 1 & 5 & -3 \\ -2 & -3 & 1 \\ -1 & 2 & -2 \end{bmatrix} = 1\cdot\det\begin{bmatrix} -3 & 1 \\ 2 & -2 \end{bmatrix} - 5\cdot\det\begin{bmatrix} -2 & 1 \\ -1 & -2 \end{bmatrix}$

$+(-3)\cdot\det\begin{bmatrix} -2 & -3 \\ -1 & 2 \end{bmatrix}$

$= 1\cdot 4 - 5\cdot 5 + (-3)\cdot(-7) = \boxed{0}$

(c) $\det\begin{bmatrix} 1 & 0 & -1 & 3 \\ 0 & -2 & 1 & 2 \\ -1 & 0 & -2 & 0 \\ 3 & 2 & 0 & -1 \end{bmatrix} = 1\cdot\det\begin{bmatrix} -2 & 1 & 2 \\ 0 & -2 & 0 \\ 2 & 0 & -1 \end{bmatrix} + (-1)\det\begin{bmatrix} 0 & -2 & 2 \\ -1 & 0 & 0 \\ 3 & 2 & -1 \end{bmatrix}$

$-3\cdot\det\begin{bmatrix} 0 & -2 & 1 \\ -1 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix}$

$= 1\cdot 4 + (-1)\cdot(-2) - 3\cdot 10 = \boxed{-24}$

12.3 Proof: According to (12.8), the formula for computing A_{44} has totally 24 (4×6) terms. However, by the basket weaving scheme for a 4×4 matrix similar to (12.10), we only get 8 terms. So $\det(A_{4\times 4})$ cannot be computed by the basket weaving scheme.

12.4. (a) $\det\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} = \det\begin{bmatrix} 2 & 0 \\ -3 & 4 \end{bmatrix} + \det\begin{bmatrix} 0 & -1 \\ -3 & 4 \end{bmatrix} = 8 - 3 = 5$ (property 2)

(b) $\det\begin{bmatrix} 1 & 5 & -3 \\ -2 & -3 & 1 \\ -1 & 2 & -2 \end{bmatrix} \xlongequal{\text{Property 5}} \det\begin{bmatrix} 1 & 5 & -3 \\ -1 & 2 & -2 \\ -1 & 2 & -2 \end{bmatrix} \xlongequal{\text{Property 4}} 0$

(c) $\det\begin{bmatrix} 1 & 0 & -1 & 3 \\ 0 & -2 & 1 & 2 \\ -1 & 0 & -2 & 0 \\ 3 & 2 & 0 & -1 \end{bmatrix} \xlongequal{\text{Property 5}} \det\begin{bmatrix} 1 & 0 & -1 & 3 \\ 0 & -2 & 1 & 2 \\ 0 & 0 & -3 & 3 \\ 0 & 2 & 3 & -10 \end{bmatrix} \xlongequal{\text{Property 5}} \det\begin{bmatrix} 1 & 0 & -1 & 3 \\ 0 & -2 & 1 & 2 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 4 & -8 \end{bmatrix}$

$\xlongequal{\text{Property 5}} \det\begin{bmatrix} 1 & 0 & -1 & 3 \\ 0 & -2 & 1 & 2 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 0 & -4 \end{bmatrix} \xlongequal{\text{Property 7}} -24$

12.5. Trivial.

12.6. (a) $\det\begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & -1 \\ 1 & 0 & 3 \end{bmatrix} = -(-1)\det\begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} = 4.$

(b). $\det\begin{bmatrix} 0 & 2 & 0 & -3 \\ 1 & -1 & 4 & 9 \\ 0 & 1 & 0 & 0 \\ 5 & 3 & -2 & 7 \end{bmatrix} = -1\cdot\det\begin{bmatrix} 0 & 0 & -3 \\ 1 & 4 & 9 \\ 5 & -2 & 7 \end{bmatrix}$

$= (-1)\cdot\left\{(-3)\det\begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}\right)$

$= (-1)(-3)(-22)$

$= -66$

12.7. 12.2 (b) is singular.

12.8. $AA^{-1} = I \Rightarrow$ By property 8, we have $\det I = \det(A)\det(A^{-1})$

$\Rightarrow \det(A) = \dfrac{1}{\det(A^{-1})}$

12.9. Yes. Since

$\det(AB) = \det(A)\det(B) = \det(B)\det(A) = \det(BA)$